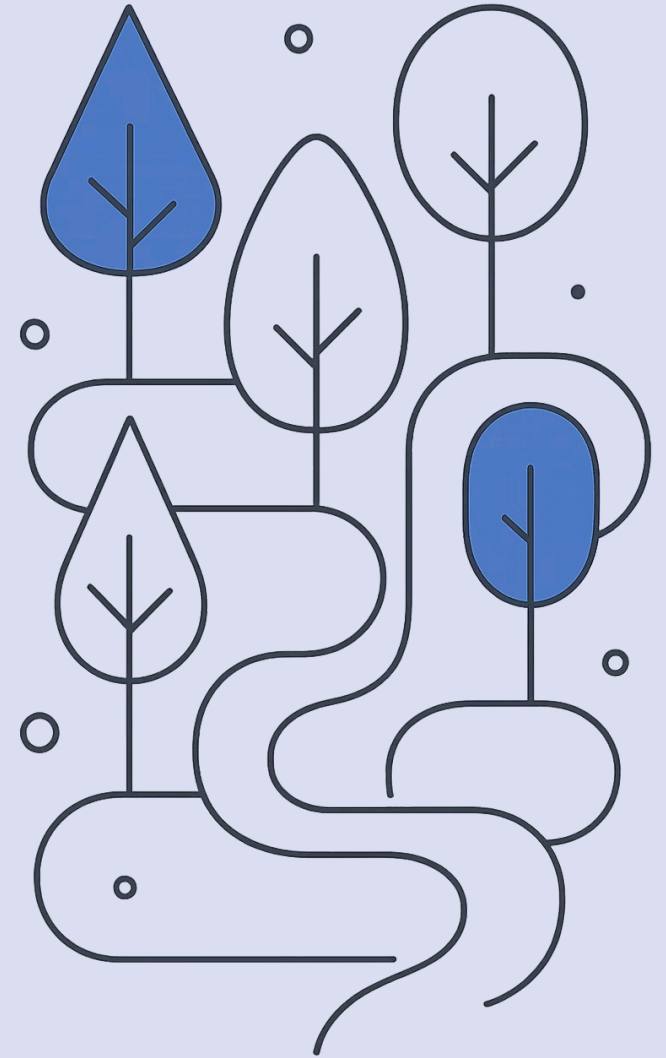


Decision Tree & Random Forest Regression

Moving beyond straight-line thinking — from formulas to questions, conditions, and decisions.

MACHINE LEARNING

REGRESSION MODELS



Why Tree-Based Models?

Linear Regression fits one formula:

 $\text{Price} = b_0 + b_1 \times \text{Area} + b_2 \times \text{Bedrooms} + b_3 \times \text{Age} + b_4 \times \text{Distance}$

Useful when the relationship is roughly straight-line — but real life works more like rules.

Prime + Big Area

→ Very high price

Far + Old House

→ Low price

Medium + Near
Metro

→ Good price

New + Far from City

→ Medium price

Why Is It Called a Tree?



The structure resembles a tree — branching from a root question down to leaf nodes.

- ❏ Leaf nodes are the final prediction points at the end of each branch.

House Price Example Data

Area	Bedrooms	Age	Distance	Price (₹L)
900	2	10	12	42
1500	3	3	5	82
2000	4	2	4	120
2500	4	1	2	155

Area > 1900 & Distance ≤ 4
→ High price

Area ≤ 1200 & Age > 5
→ Low price

Area 1400–1900
→ Medium price

Linear Regression vs Decision Tree Thinking

Linear Regression

Fit one best mathematical formula.

①
$$\text{Price} = 10 + 0.04 \times \text{Area} + 5 \times \text{Bedrooms} - 1 \times \text{Age} - 2 \times \text{Distance}$$

Decision Tree

Ask questions and split data into groups.

- Is area greater than 1800?
- Is distance less than 5?
- Is age less than 5?

The Real Estate Agent Analogy

Linear Regression Agent

- Every extra sqft adds ₹4,000
- Every extra bedroom adds ₹5L
- Every extra km reduces ₹2L

Decision Tree Agent

- First check location
- Then check area
- Then check age & distance
- Estimate based on conditions

Both are useful — they just think differently.



Decision Tree: Prediction Walkthrough

Learned Tree Rules

Area > 1800 &
Distance < 5

→ ₹130 lakh

Area > 1800 &
Distance ≥ 5

→ ₹105 lakh

Area ≤ 1800 & Age <
5

→ ₹85 lakh

Area ≤ 1800 & Age ≥
5

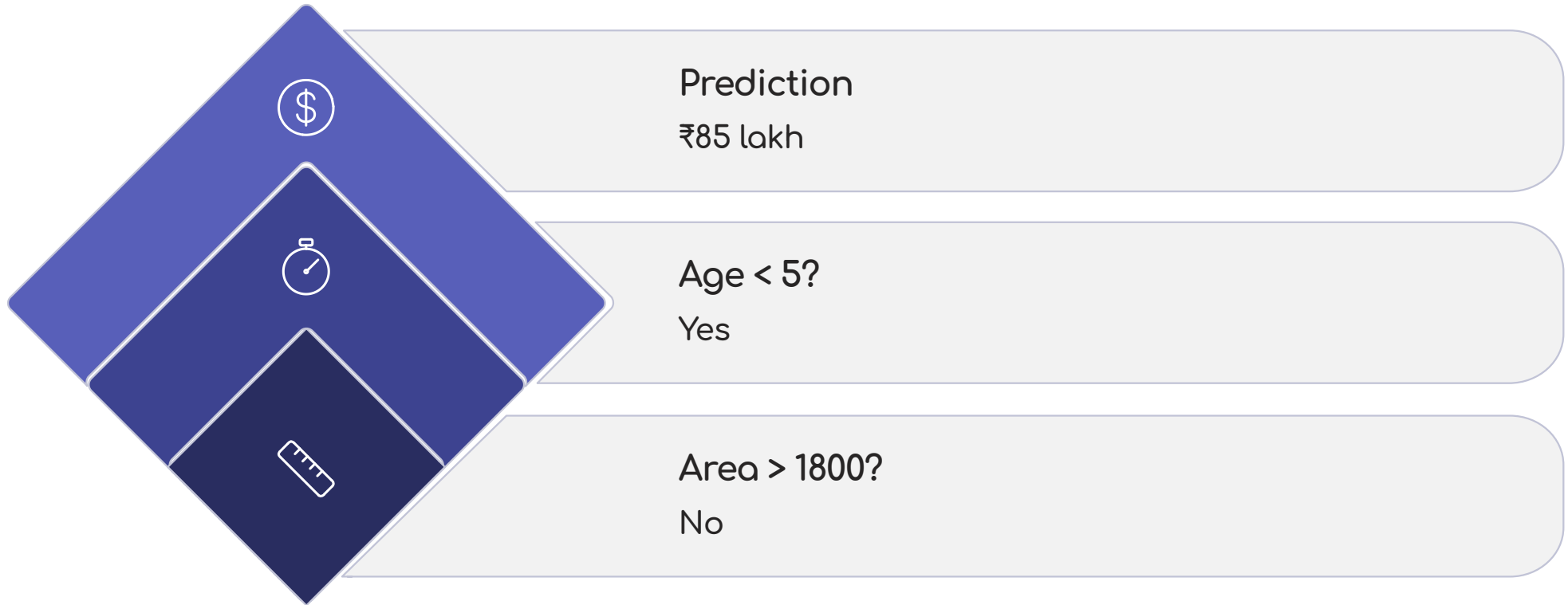
→ ₹55 lakh

New House Input

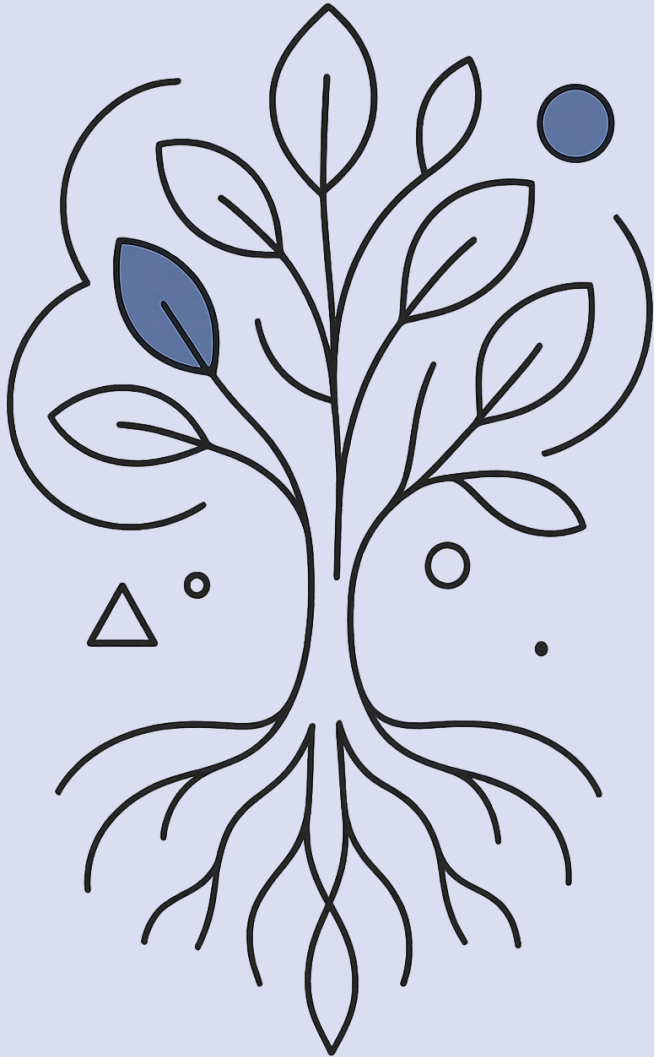
Area = 2000 | Distance = 4 | Age = 3

✔ Area > 1800? ✔ Yes
Distance < 5? ✔ Yes
Prediction = ₹130 lakh

Another Prediction Example



New house: Area = 1500, Distance = 6, Age = 3. The tree follows question paths until it reaches a final prediction at a leaf node.



Strengths of Decision Tree Regression



Easy to Understand

Visual, interpretable structure



Non-Linear Patterns

Captures complex relationships



Rule-Based Behavior

Mirrors real-world conditions



No Feature Scaling

Works without normalization

The Major Weakness: Overfitting

Overfit

The model memorizes training data and performs poorly on new data.

Too Many Tiny Rules

Area=1500, Age=3,
Dist=5 → ₹82L

Area=1510, Age=3,
Dist=5.2 → ₹83L

Area=1520, Age=4, Dist=5.1 → ₹81L



Hyper-specific rules make the tree weak on unseen data.

Decision Tree Regression: Code

```
import pandas as pd
from sklearn.tree import DecisionTreeRegressor

data = {
    "area": [900, 1000, 1500, 1800, 2000, 2200, 2500],
    "bedrooms": [2, 2, 3, 3, 4, 4, 4],
    "age": [10, 5, 3, 4, 2, 2, 1],
    "distance": [12, 8, 5, 6, 4, 3, 2],
    "price": [42, 55, 82, 95, 120, 130, 155]
}
df = pd.DataFrame(data)
X = df[["area", "bedrooms", "age", "distance"]]
y = df["price"]

model = DecisionTreeRegressor(random_state=42)
model.fit(X, y)
```

Decision Tree: Prediction Code

```
new_house = pd.DataFrame({  
    "area": [1700],  
    "bedrooms": [3],  
    "age": [4],  
    "distance": [6]  
})  
  
prediction = model.predict(new_house)  
print("Predicted price:", prediction[0], "lakhs")
```

 DecisionTreeRegressor = Decision Tree model for predicting numbers (regression).

What Is Random Forest Regression?

Random Forest Regression combines many Decision Trees and averages their predictions.

Problem	Solution
One decision tree can overfit	Let many trees work together
Result	
Average all tree predictions	



Why "Random" Forest?

The Naming Logic

- Forest = many trees
- Random = each tree trained slightly differently

Decision Tree = one tree

Random Forest = many decision trees

₹85L

Tree 1

₹88L

Tree 2

₹82L

Tree 3

₹86L

Final Avg

The Expert Panel Analogy

One Agent (Decision Tree)

⚠ One agent may be biased → ₹90 lakh

Ten Agents (Random Forest)

₹85L, ₹88L, ₹92L, ₹86L, ₹89L...

✅ Take the average → more reliable estimate

Decision Tree = one expert | Random Forest = group of experts



Random Forest: Prediction Example

Tree	Prediction
Tree 1	₹80 lakh
Tree 2	₹85 lakh
Tree 3	₹82 lakh
Tree 4	₹86 lakh
Tree 5	₹87 lakh

✔ $(80 + 85 + 82 + 86 + 87) / 5 = ₹84 \text{ lakh}$ — Final Prediction

Why Random Forest Is Powerful



Better Accuracy

Usually outperforms a single decision tree



Reduces Overfitting

Averaging many trees smooths out errors



Feature Importance

Reveals which features matter most



Handles Complexity

Works well with many features & non-linear patterns

Random Forest Regression: Code

```
import pandas as pd
from sklearn.ensemble import RandomForestRegressor

data = {
    "area": [900, 1000, 1500, 1800, 2000, 2200, 2500],
    "bedrooms": [2, 2, 3, 3, 4, 4, 4],
    "age": [10, 5, 3, 4, 2, 2, 1],
    "distance": [12, 8, 5, 6, 4, 3, 2],
    "price": [42, 55, 82, 95, 120, 130, 155]
}
df = pd.DataFrame(data)
X = df[["area", "bedrooms", "age", "distance"]]
y = df["price"]

model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X, y)
```

Random Forest: Prediction Code

```
new_house = pd.DataFrame({  
    "area": [1700],  
    "bedrooms": [3],  
    "age": [4],  
    "distance": [6]  
})  
  
prediction = model.predict(new_house)  
print("Predicted price:", prediction[0], "lakhs")
```

 `n_estimators=100` → Build 100 decision trees and average their predictions.

Decision Tree vs Random Forest

Point	Decision Tree	Random Forest
Model Type	One tree	Many trees
Prediction	Single tree output	Average of all trees
Interpretability	Easy	Medium
Overfitting Risk	Higher	Lower
Accuracy	Good	Often better
Stability	Less stable	More stable

Linear Regression vs Tree Models

Point	Linear Regression	Decision Tree / Random Forest
Relationship	Straight-line assumption	Rule-based splits
Non-Linear Patterns	Limited	Better
Explainability	Formula-based	Tree/rule-based
Overfitting Risk	Usually lower	Tree can overfit
Real-World Performance	Good baseline	Often stronger

When to Use Linear Regression

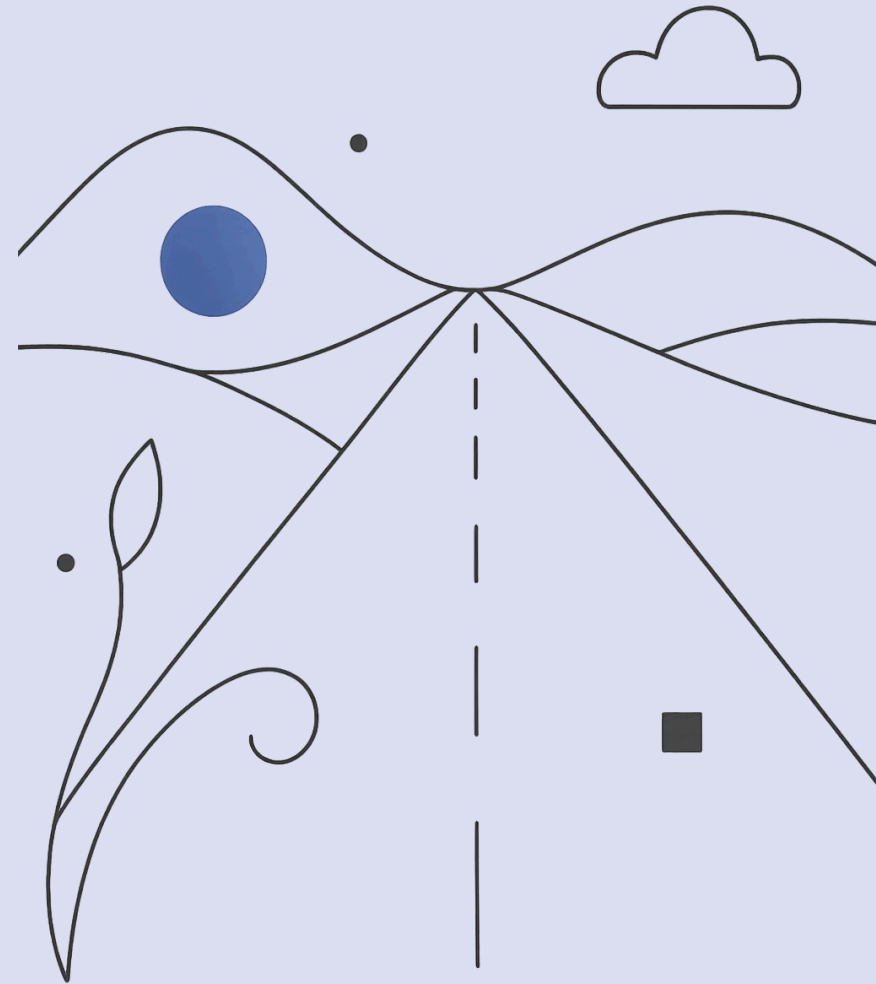
☐ Simple, explainable formula needed
Relationship looks roughly linear

☐ Understand feature direction
Data is not too complex

Experience → Salary

Area → Price

Ad Spend → Sales



When to Use Decision Tree

Best Fit When:

- You want rule-based explanation
- Data has non-linear conditions
- You want simple visual decision logic

Example Use Case

 If income > X and credit score > Y → loan amount estimate

Decision Tree shines when business logic feels like conditions and branches.

When to Use Random Forest

- Stronger performance needed
Many features, complex relationships
- Reduce overfitting
More stable than a single tree
- Real-world examples
House prices, sales forecasting, delivery time prediction with traffic, weather & distance



Common Confusion #1

Is Decision Tree only for classification?

✅ No! Decision Trees work for both tasks.

The algorithm structure is the same — only the output type differs.

DecisionTreeClassifier

Predicts a category

DecisionTreeRegressor

Predicts a number

Common Confusion #2

Does Random Forest always give perfect results?

 No. Random Forest is powerful, but not magic.

Good Data

Garbage in, garbage out

Relevant Features

Right inputs matter

Enough Examples

Sufficient training data

Proper Evaluation

Correct train-test split

Common Confusion #3

Is Random Forest always better than Linear Regression?

Simple Pattern

Linear Regression may work very well — straightforward, interpretable, fast.

Complex Pattern

Random Forest often helps when relationships are non-linear and multi-dimensional.

 Simple pattern → Linear Regression may be enough | Complex pattern → Random Forest may help

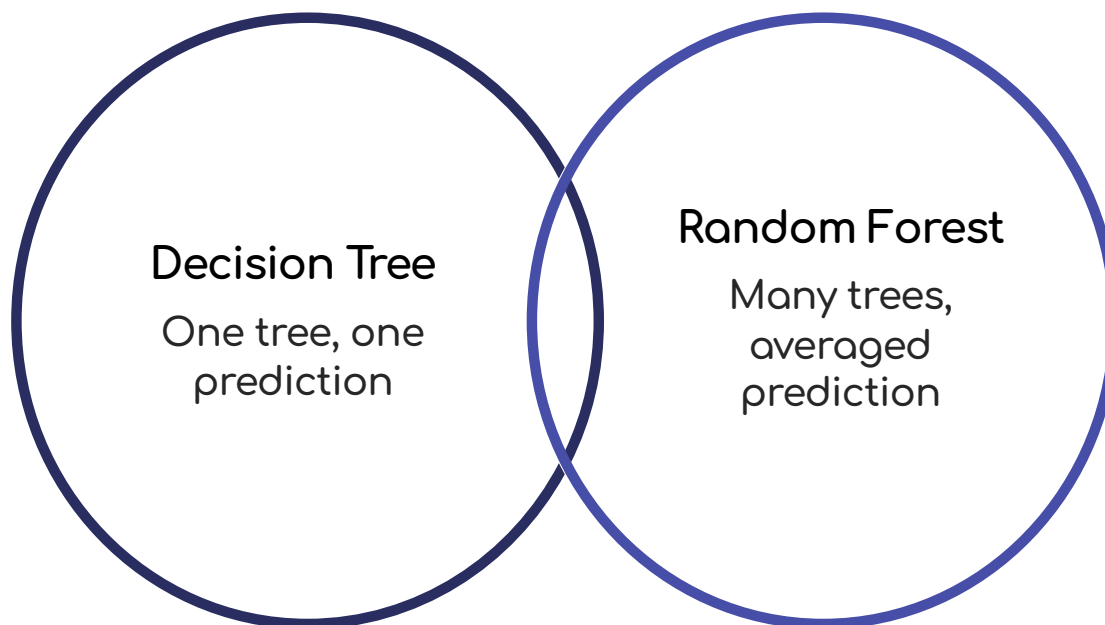
Final Summary

Decision Tree Regression

Predicts a number using decision rules — ask questions like $\text{area} > 1800$, $\text{distance} < 5$, $\text{age} < 5$.

Random Forest Regression

Many decision trees together — ask all trees and average their answers for a more robust prediction.



Memory Hook

1

Linear Regression
Formula thinking

2

Decision Tree
Flowchart thinking

3

Random Forest
Group of flowcharts



Next Topic: Regression Evaluation Metrics — MAE, MSE, RMSE, and R^2 Score